

Problem Statement



Problem Statement - Background



Blood Warriors

- Blood Warriors is a foundation connecting voluntary blood donors with Thalassemia patients across India. With over 1,00,000 patients requiring up to 500 to 700 transfusions in a lifetime, the demand for reliable, recurring, and compatible blood is constant. Their Blood Bridge initiative and donor network are at the heart of making this happen.

The Problem

- Today, matching donors to patients, coordinating requests, and keeping the ecosystem engaged relies heavily on manual effort. As Blood Warriors scales across cities and partner organizations, this becomes harder to sustain.

AI-Enabled Care Coordination and Access

- How might we make it easier for different people involved in care to stay connected and informed at the right time?
- What could more intuitive, adaptive interactions between individuals and systems look like across languages and contexts?
- How might systems remember and respond appropriately over repeated interactions?

Problem Statement - Background



Smart Matching and Resource Allocation

- How can we better identify and prioritize the right people or resources when needs arise?
What signals or patterns could be used to anticipate availability or willingness to participate?

Engagement and Continuity

- How might we encourage continued participation and build long term engagement among contributors?
- What kinds of feedback loops or interactions could make the system feel more responsive and human?

Operating at Scale

- What would it take for such a system to function effectively across a large, diverse population?
- How might it adapt to variations in language, access, and infrastructure without losing efficiency?
- What considerations come into play when expanding to support many users and regions simultaneously?

Problem Statement



Build an autonomous, AI-powered blood support network that can coordinate requests, engage the right participants, and manage interactions in real time with minimal manual effort.

What the System Should Be Able to Do

- Handle multiple workflows through a unified intelligent AI layer
- Automate outreach, follow-ups, and escalations for donors
- Track and interpret donor responses to guide next steps
- React to real-time events and updates
- Enable conversational interactions with memory
- Systems must self manage improvement steps and protocols via failure learning
- Provide admins with dashboards and insights
- Ensure consent-aware, responsible data usage and compliant systems

Infrastructure



Infrastructure - AWS

The following **AWS Services** are authorized for use alongside **GitHub** integrations:

Compute & Web

- S3 + CloudFront / Amplify
- AWS Cognito
- AWS App Runner
- Lambda + API Gateway
- Fargate / Amazon EKS

Events & Streams

- Amazon SQS / SNS / SES
- Amazon EventBridge
- IAM / CloudWatch
- Kinesis Data Streams
- AWS Glue

Databases

- Amazon DynamoDB
- RDS (Postgres / Aurora)
- Amazon Redshift
- Amazon Athena
- OpenSearch Service

Security & Ops

- Secrets Manager / Parameter Store
- WAF / AWS Shield
- AWS X-Ray
- Managed Airflow (MWAA)

AI & Machine Learning

- Amazon Bedrock
- Amazon SageMaker
- Amazon Lex
- SageMaker Feature Store

DevOps

- AWS CodeBuild
- AWS CodePipeline
- AWS CodeCommit
- GitHub Actions

We will send a soft notification when usage goes beyond \$30, followed by another notification if it exceeds \$40. Participants will be asked to downgrade services (e.g. Save work on GitHub) upon exceeding \$40.

Recommended Tech Stack

1

Full Stack (User Interaction Layer)

Recommended:

Front-end:
React.js (web UI)
Back-end:
Python (Fast API / Flask), REST APIs + Async processing

2

Data Engineering (Pipelines and Storage)

Recommended:

Database: AWS Databases (RDS, S3, Aurora, DynamoDB)
ETL pipelines services via AWS(Glue, Kinesis)

3

Data Science & AI (Intelligence Layer)

Recommended:

ML: Matching / Ranking models, NLP, Predictions Layers(SageMaker)
AI: Conversational AI(BedRock)

4

Automation Systems (Orchestration Layer)

Recommended: AWS Step function, AWS Lambda, API Gateway

5

Deployment and Monitoring

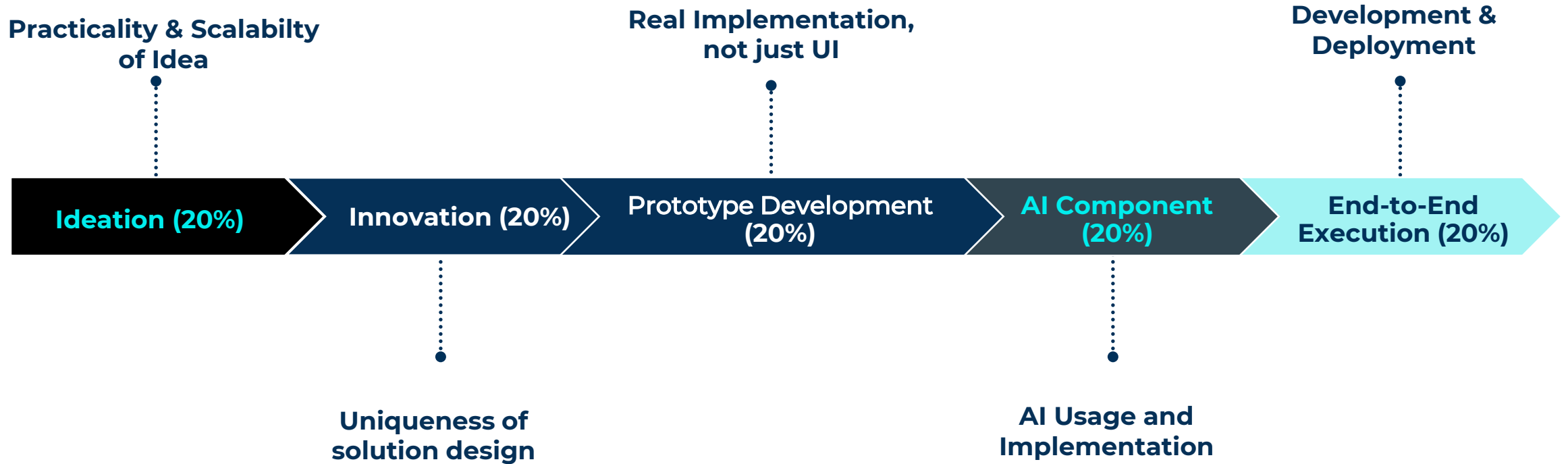
Recommended:

AWS EC2, CloudWatch
Optional: ECS + AWS CodePipeline



Judging Criteria

Evaluation Criteria



Hackathon Flow



AI For Good 2.0 : Overall SCHEDULE

Two-day schedule at a glance

DAY 1

START

11:00 AM

Kick-off

CHECKPOINT

04:00 PM

Checkpoint 1

Ice Breaker

07:00 PM

Ice Breaker Session 1

DAY 2

CHECKPOINT

09:00 AM

Checkpoint 2

DEADLINE

11:00 AM

Submission

JUDGEMENT

01:30 PM

Jury Session

Day 1 · Detailed Schedule

8:00 – 10:00 AM	Registrations & Check-in	3:30 – 4:00 PM	Hi-Tea & Snacks
8:30 – 9:30 AM	Breakfast	4:00 PM Onwards	1st Checkpoint & Mentorship
10:00 – 11:00 AM	Opening Ceremony, Keynote, Ground rules	4:00 – 7:00 PM	Hack Hours
11:00 AM	Hackathon Kick-off	7:00 – 7:30 PM	Ice Breaker Session 1
11:00 AM – 1:00 PM	Hack Hours	7:30 – 9:00 PM	Hack Hours
1:00 PM	Lunch	9:00 PM	Dinner
2:00 – 3:30 PM	Hack Hours	10:00 – 11:59 PM	Hack Hours

Day 2 · Detailed Schedule

12:00 – 12:30 AM	Icebreaker Session 2	12:00 PM	Top 15 Team Announcements
12:30 – 8:00 AM	Hack Hours	12:00 – 1:00 PM	Lunch
8:00 – 9:00 AM	Breakfast	1:00 – 4:30 PM	Jury Sessions
9:00 AM Onwards	2nd Checkpoint & Mentorship	4:30 – 6:00 PM	Closing Keynote Address
9:00 – 11:00 AM	Hack Hours	4:30 – 6:00 PM	Announcement of Winners & Results
11:00 AM	Hackathon Ends with Submissions	4:30 – 6:00 PM	Vote of Thanks
11:00 AM – 12:00 PM	Break		